



The GeForce FX can spell all the difference between a frown and a wry smile. The effect is apparent, as modelled by Dawn, the pixie mascot of the GeForce FX

nologies that help deliver this promise of cinema-quality gaming.

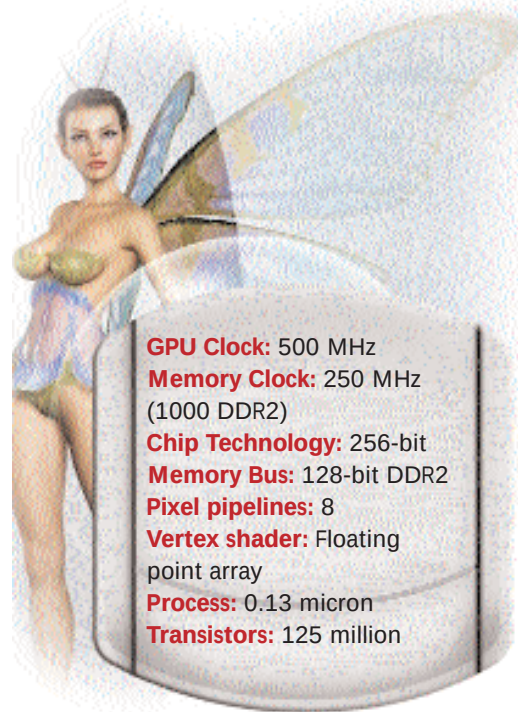
Parallel vertex shader and pixel pipelines

The vertex shader is the element that allows complex meshes and 3D objects to be created within games. In the GeForce FX, this component was created from scratch—while the GeForce 4 had two separate vertex shader units, the GeForce FX

features just a single pipeline. This singular pipeline is equipped with parallel floating point units that work together to deliver a claimed triangle processing rate of 375 million triangles per second (compared to the Radeon 9700's 325 million triangles per second). When coupled with the DirectX 9 specification, the card can process vertex shader programs while using 65,536 instructions (the GeForce4 Ti class cards uses only 128 instructions), constants and loops (primitives that are used in the creation of these programs). So what does all this translate into? While earlier cards allowed only for models with increasing polygon counts, the advanced vertex shading techniques of the GeForce FX will make even facial expressions possible.

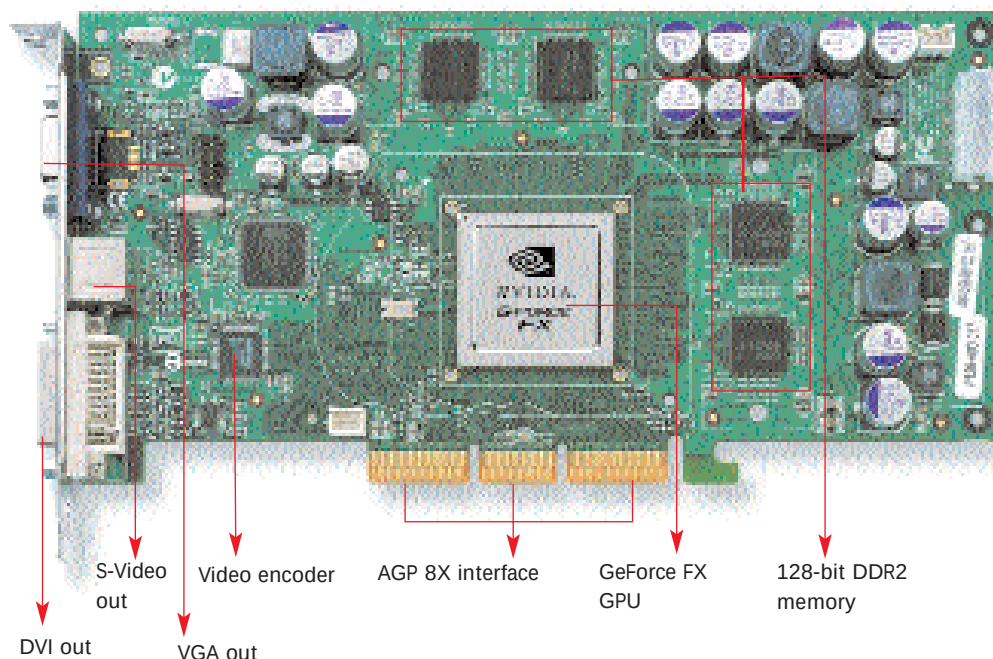
While vertex shader warrants greater flexibility and innovation in modifying meshes and therefore greater realism, the pixel shader does wonders for the textures and special effects on the surfaces of these 3D models. Everything from the mouldy rusted iron girders to the clammy, cold texture of an alien's skin will be defined using these new innovations.

The pixel shader unit of the processor in the GeForce FX has been considerably revamped to add functionality for the DirectX 9 API. Also, pixel shader programs can now use up to 1,024 instructions with up to 16 textures per pixel, allowing for very complex effects on the 3D model surfaces. Some of the new pixel operations are soft shadows on 3D models and frame-



GPU Clock: 500 MHz
Memory Clock: 250 MHz
 (1000 DDR2)
Chip Technology: 256-bit
Memory Bus: 128-bit DDR2
Pixel pipelines: 8
Vertex shader: Floating point array
Process: 0.13 micron
Transistors: 125 million

buffer post processing effects such as blurs and halos (from streetlamps and car headlights). Programmers can also assign functions to certain textures to call upon specific effects and actions. Volumetric rendering can be implemented through algorithms for generating effects such as smoke, fire, fur, grass, etc. Complex lighting effects can be achieved over a single pass as opposed to using up many clock cycles for rendering a given light effect. Finally, the pixel shader also features 64 temporary registers (compared to 12 for the Radeon 9700), which are used for storing complex instructions that are sent to the graphics processor.



DDR2 memory

Another big advantage is the support for DDR2 memory. Operating at a blistering 500 MHz frequency on a 128-bit data bus, this interface offers a 16 GBps bandwidth. Even though the memory width is lower than that of the Radeon 9700, the effective bandwidth is higher for two reasons: the effective frequency is higher with the DDR2 memory and every bit of data that comes out of the rendering pipeline is compressed in the hardware before being sent to the memory. On an average, nVidia states that there is a 4:1 compression ratio that occurs and therefore, the resultant memory bandwidth of the card is effectively raised to 48 GBps—that's the contents of an entire